



Cloud Computing

LAB #02 (Hands-On Git & Version Control Lab)

SUBMITTED BY:

NAME: Manahil Habib

SECTION: B

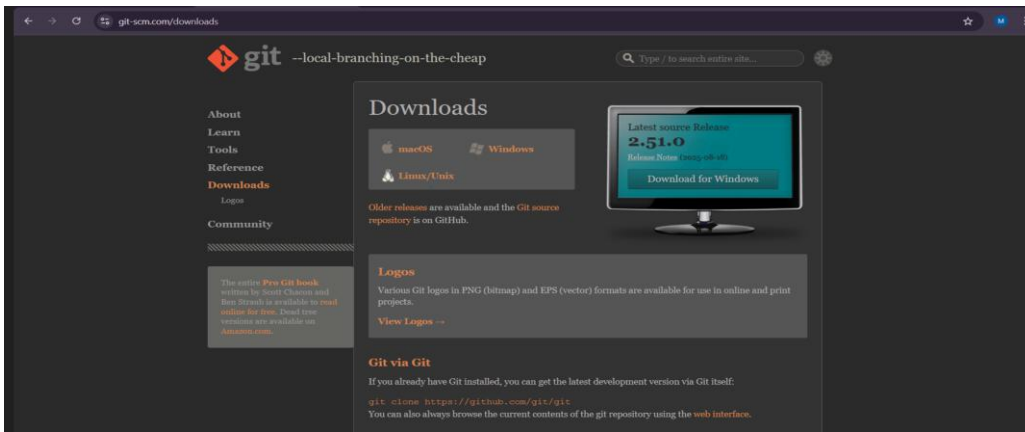
REG NO: 2023-BSE-042

SUBMITTED TO: Engr. M. Shoaib

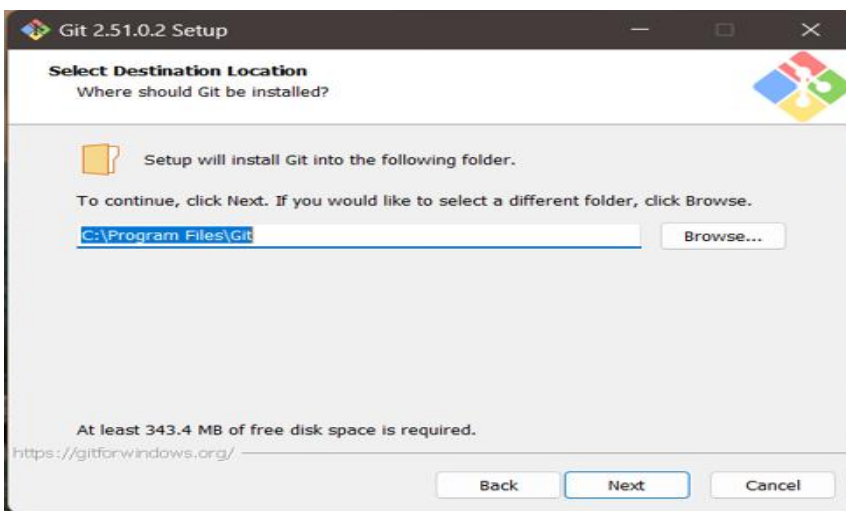
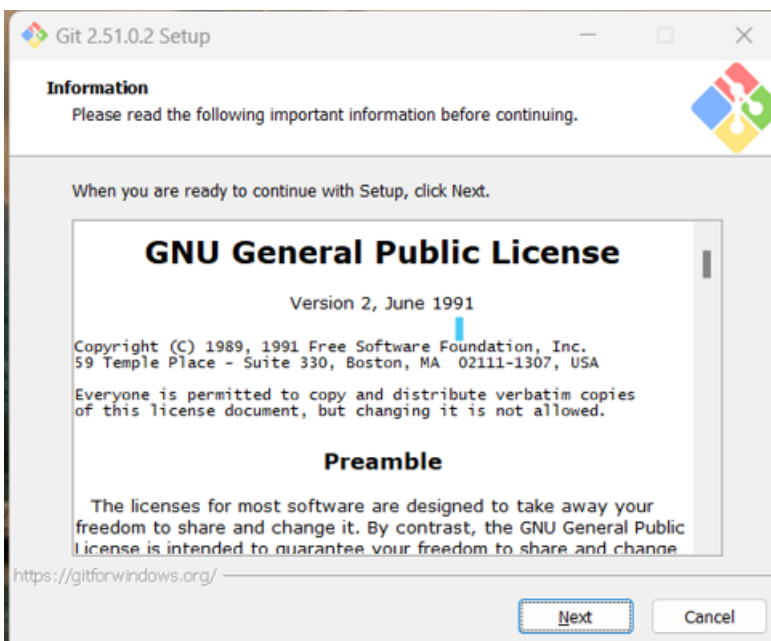
Getting Started

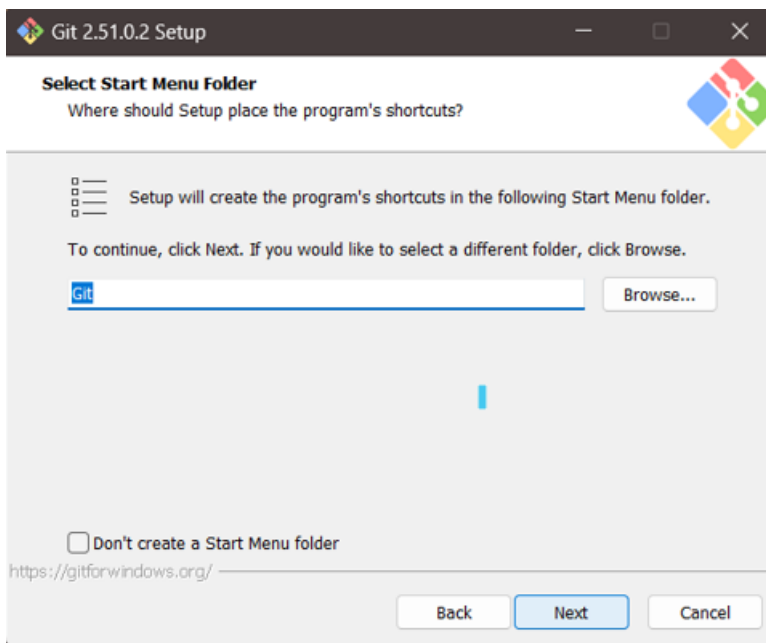
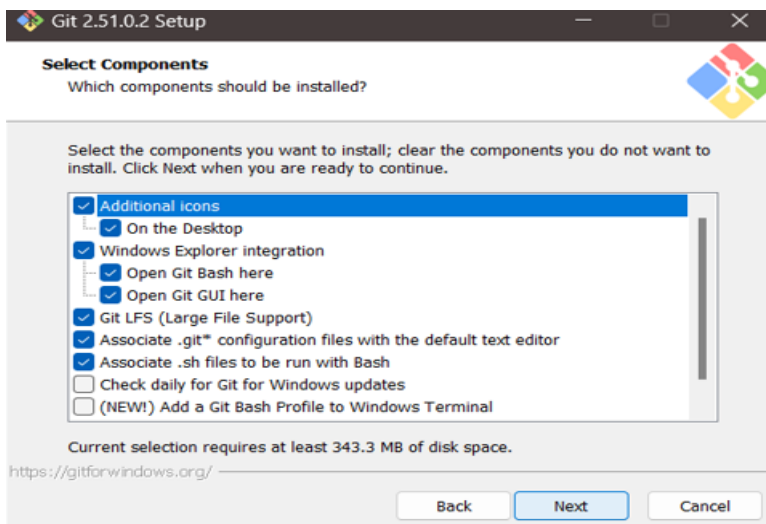
Install Git on your PC.

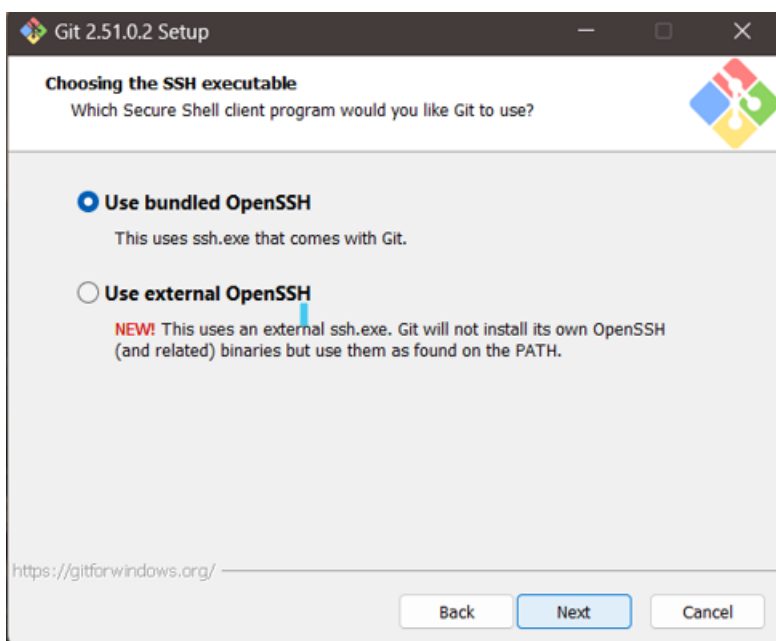
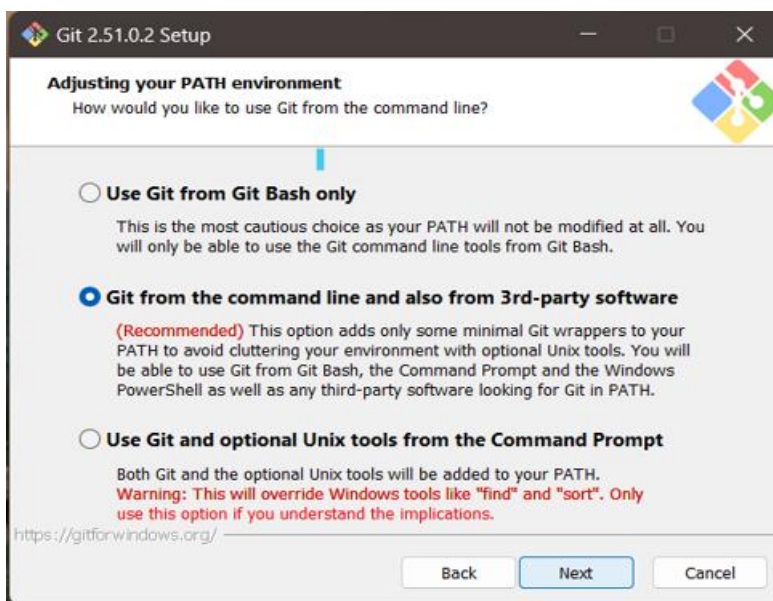
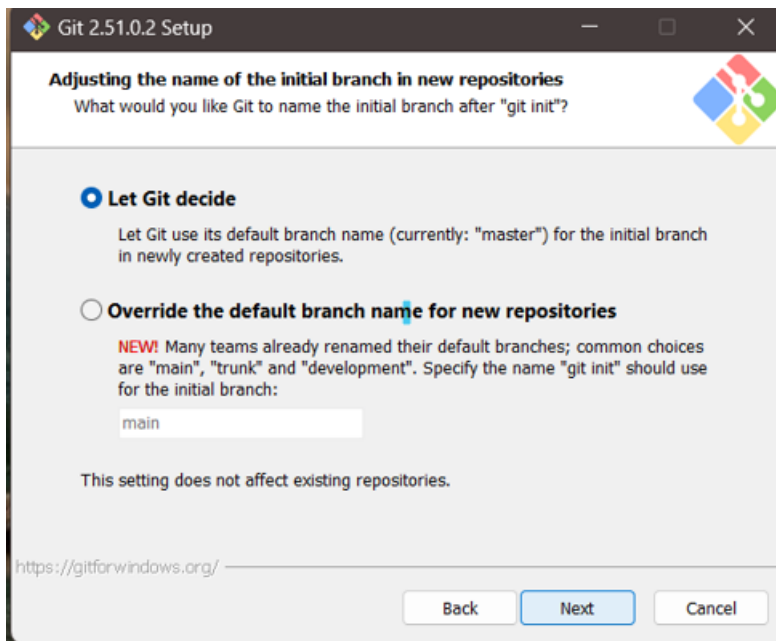
From google, download git for winndows

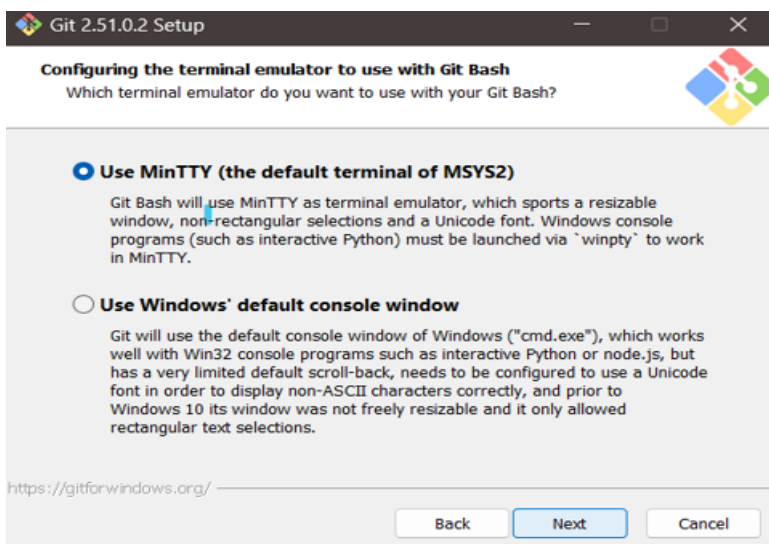
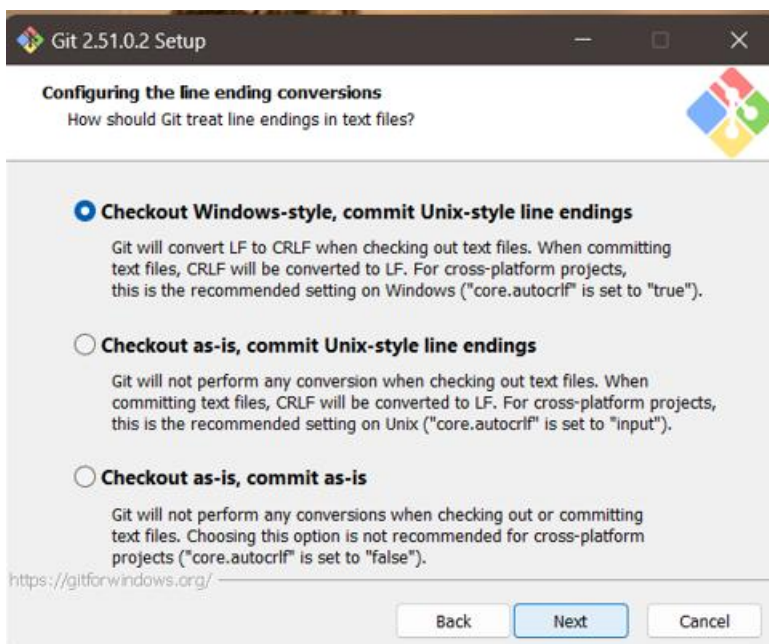


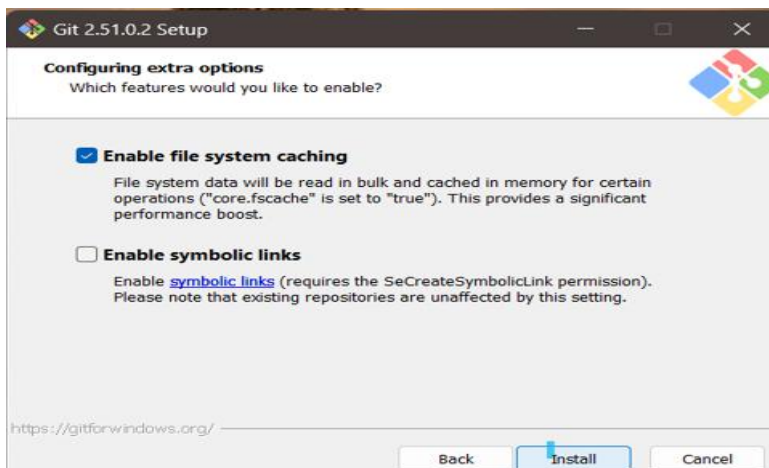
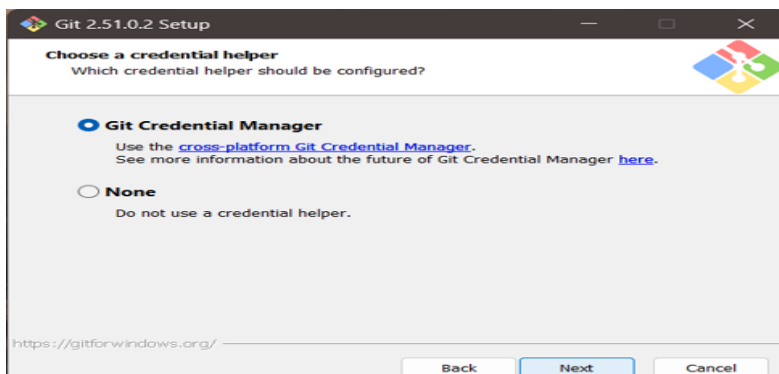
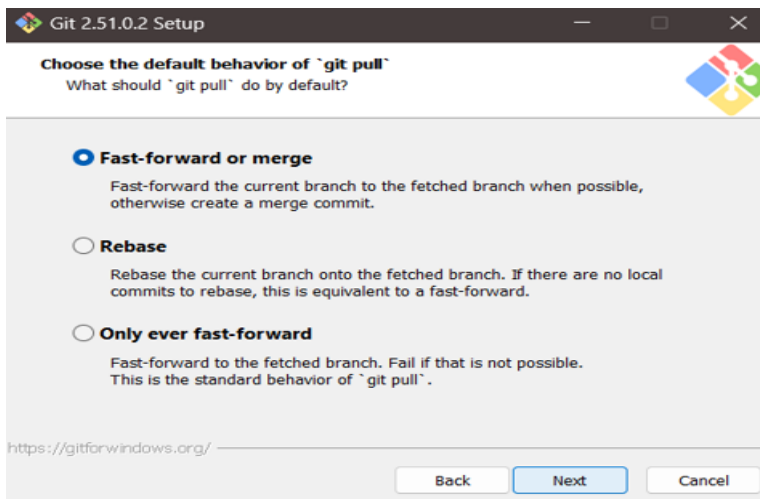
check the desired box and click on “next”



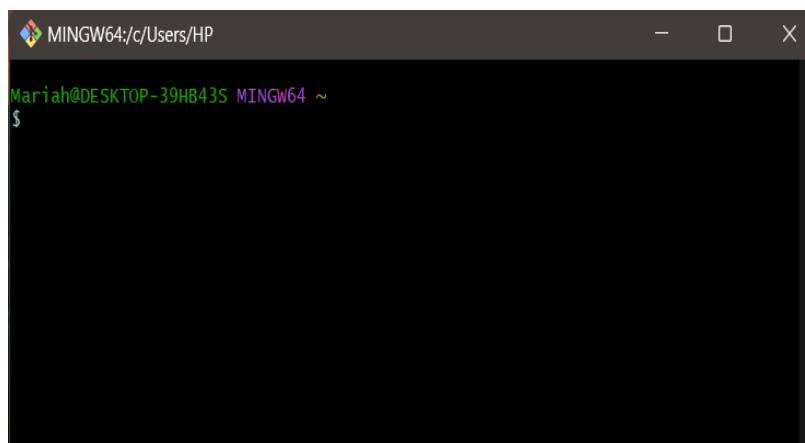








Installation Completed.



TASK NO 01:

Create Private GitHub Repository

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk ().*

1

General

Owner *

manahilhabib031

Repository name *

Lab2

✓ Lab2 is available.

Great repository names are short and memorable. How about **fuzzy-waddle**?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

Private

Add README

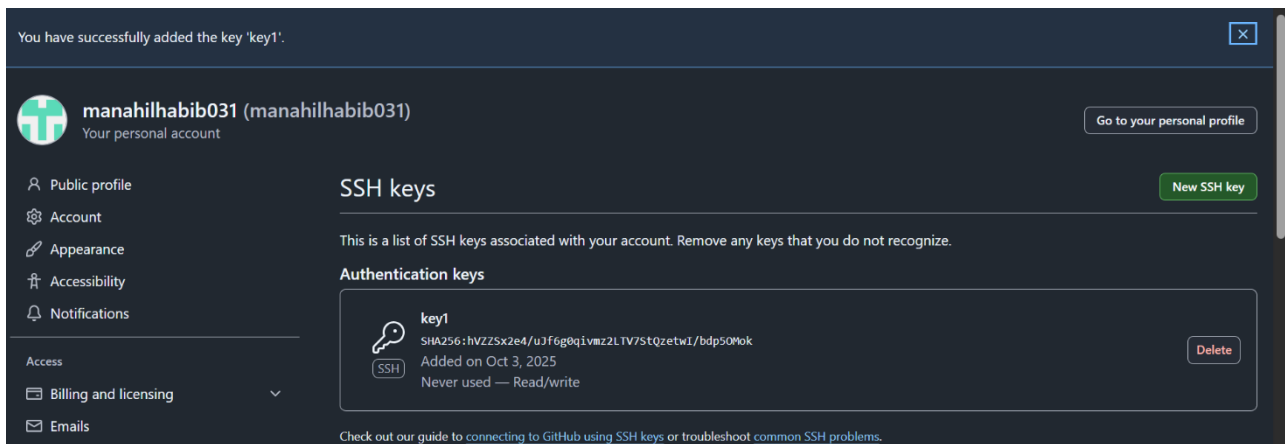
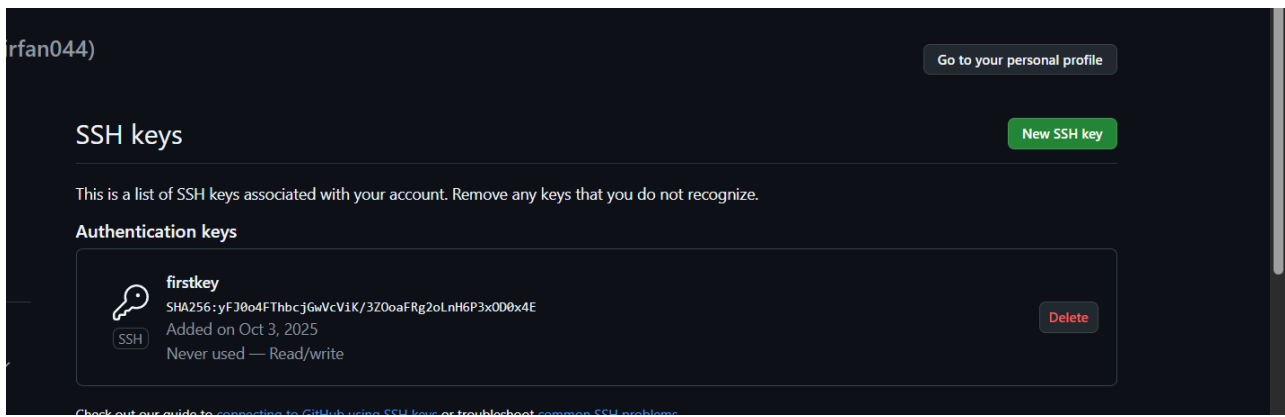
Off

READMEs can be used as longer descriptions. [About READMEs](#)

TASK NO 02:

Connect Repository via SSH

```
PS C:\Users\HABIB TARIQ> ssh-keygen -t ed25519 -C "22-22411-042@se.fjwu.edu.pk"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\HABIB TARIQ\.ssh/id_ed25519):
Created directory 'C:\\Users\\HABIB TARIQ\\.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\HABIB TARIQ\.ssh/id_ed25519
Your public key has been saved in C:\Users\HABIB TARIQ\.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:hVZZSx2e4/uJf6g0qivmz2LTV7StQzetwI/bdp50Mok 22-22411-042@se.fjwu.edu.pk
The key's randomart image is:
+--[ED25519 256]--+
|      .oo... |
|     o.. o.. |
|    o . . +  |
|   . . . . . |
|  S  o o.. |
|     B =.. |
|    EoX.= |
|   *.. .o+oX.= |
|  +..==+. o*+B+ |
+-----[SHA256]-----+
PS C:\Users\HABIB TARIQ>
```

```
$ git clone https://github.com/manahilhabib031/Lab2.git
Cloning into 'Lab2'...
warning: You appear to have cloned an empty repository.

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~
$
```

TASK NO 03:

Configure Git Username and Email

1:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~
$ git config --global user.name "manahilhabib031"

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~
$ git config --global user.email "23-22411-042@se.fjwu.edu.pk"
```

2:


```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=manahilhabib031
```

TASK NO 04:

Explore the .git Folder

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~
$ cd Lab2

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ ls -a
./ ../ .git/

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ ls -a .git
./ ../ HEAD config description hooks/ info/ objects/ refs/

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
```

g

TASK NO 05:

Local Repository Management

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ rm -rf .git

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2
$
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2
$ git init
Initialized empty Git repository in C:/Users/HABIB TARIQ/Lab2/.git/
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (master)
```

```
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=manahilhabib031
user.email=23-22411-042@se.fjwu.edu.pk
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (master)
```

```
$ git commit -m "Initial commit"
[master (root-commit) 25d4234] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (master)
```

```
$ git remote add origin https://github.com/manahilhabib031/Lab2.git
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (master)
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (master)
```

```
$ git branch -M main
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
```

```
$ git remote add origin https://github.com/manahilhabib031/Lab2.git
error: remote origin already exists.
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
```

```
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 243 bytes | 121.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

TASK NO 06:

File Status & Staging

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ echo "This is my note" > notes.txt

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

nothing added to commit but untracked files present (use "git add" to track)

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git add notes.txt
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git commit -m "Add notes.txt"
[main 84711d3] Add notes.txt
1 file changed, 1 insertion(+)
create mode 100644 notes.txt

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 100.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/Lab2.git
25d4234..84711d3  main -> main
```

Updated:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ echo "this is updated nots" >notes.txt

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   notes.txt

no changes added to commit (use "git add" and/or "git commit -a")

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git commit -m "Add notes.txt"
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   notes.txt

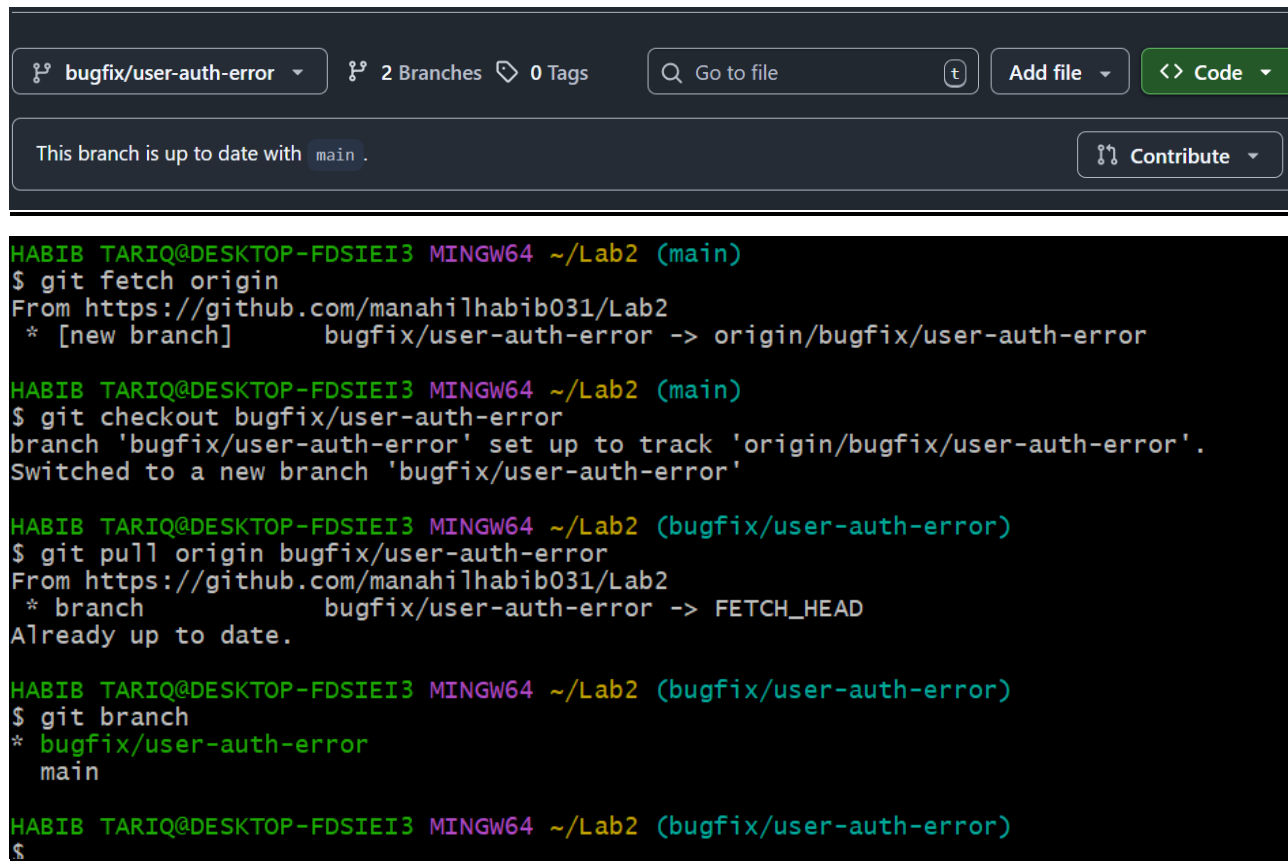
no changes added to commit (use "git add" and/or "git commit -a")

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git add notes.txt
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git commit -m "updaed notes.txt"
[main cc2bf14] updaed notes.txt
1 file changed, 1 insertion(+), 1 deletion(-)
```

TASK NO 07:

Branch Creation Using GitHub GUI



The screenshot shows the GitHub web interface at the top, indicating the repository 'bugfix/user-auth-error' with 2 branches and 0 tags. Below the interface is a terminal window showing the following commands and output:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git fetch origin
From https://github.com/manahilhabib031/Lab2
* [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git checkout bugfix/user-auth-error
branch 'bugfix/user-auth-error' set up to track 'origin/bugfix/user-auth-error'.
Switched to a new branch 'bugfix/user-auth-error'

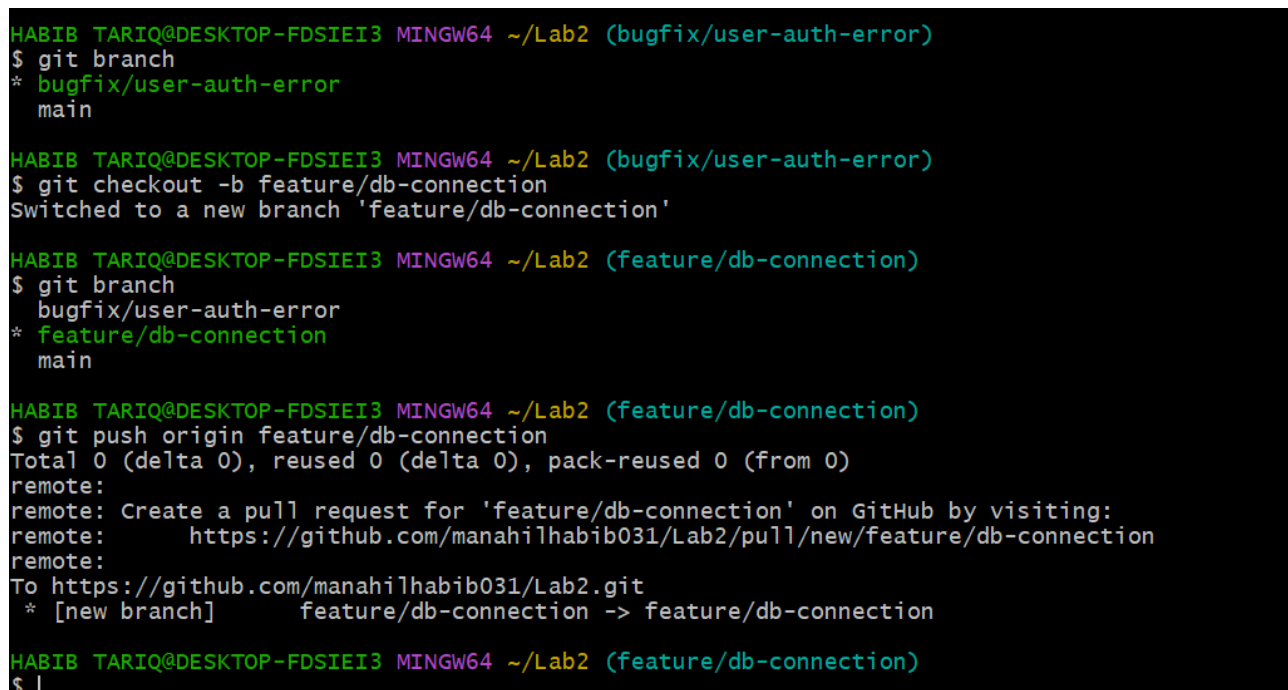
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/user-auth-error)
$ git pull origin bugfix/user-auth-error
From https://github.com/manahilhabib031/Lab2
* branch            bugfix/user-auth-error -> FETCH_HEAD
Already up to date.

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/user-auth-error)
$ git branch
* bugfix/user-auth-error
  main

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/user-auth-error)
$
```

TASK NO 08:

Branch Creation and Push Using Git Bash



The screenshot shows a terminal window with the following commands and output:

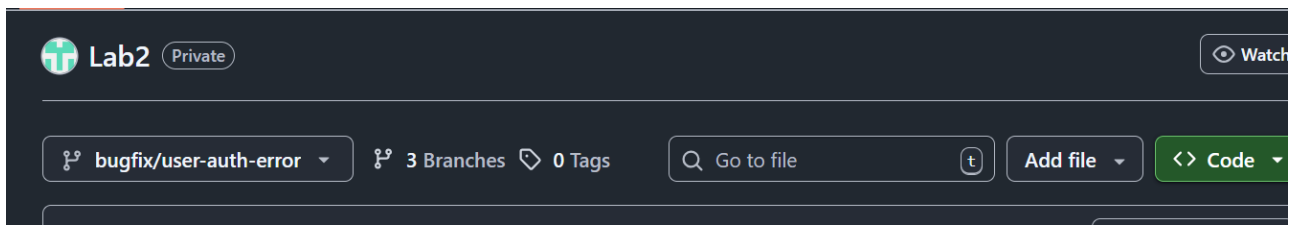
```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/user-auth-error)
$ git branch
* bugfix/user-auth-error
  main

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/user-auth-error)
$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature/db-connection)
$ git branch
  bugfix/user-auth-error
* feature/db-connection
  main

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature/db-connection)
$ git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/Lab2/pull/new/feature/db-connection
remote:
To https://github.com/manahilhabib031/Lab2.git
* [new branch]      feature/db-connection -> feature/db-connection

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature/db-connection)
$ |
```



TASK NO 09:

Branching & Merging

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-1)
$ ~/Lab2
bash: /c/Users/HABIB TARIQ/Lab2: Is a directory

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-1)
$ git checkout -b feature-1
fatal: a branch named 'feature-1' already exists

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-1)
$ echo "def new_function():" > main.py

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-1)
$ git add main.py
warning: in the working copy of 'main.py', LF will be replaced by CRLF the next time Git touches it

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-1)
$ git commit -m "Add new function to main.py"
[feature-1 8aedabe] Add new function to main.py
1 file changed, 1 insertion(+)
create mode 100644 main.py

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-1)
$ !
```

```
Merge branch 'feature-1'
# Please enter a commit message to explain why this merge is necessary,
# especially if it merges an updated upstream into a topic branch.
#
# Lines starting with '#' will be ignored, and an empty message aborts
# the commit.
~
```

```
Merge made by the 'ort' strategy.  
main.py | 1 +  
1 file changed, 1 insertion(+)  
create mode 100644 main.py
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
```

```
$ git push origin main
```

```
Enumerating objects: 10, done.
```

```
Counting objects: 100% (10/10), done.
```

```
Delta compression using up to 8 threads
```

```
Compressing objects: 100% (6/6), done.
```

```
Writing objects: 100% (8/8), 800 bytes | 72.00 KiB/s, done.
```

```
Total 8 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
```

```
remote: Resolving deltas: 100% (2/2), done.
```

```
To https://github.com/manahilhabib031/Lab2.git
```

```
84711d3..d4bc034 main -> main
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
```

```
$ git push origin feature-1
```

```
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

```
remote:
```

```
remote: Create a pull request for 'feature-1' on GitHub by visiting:
```

```
remote: https://github.com/manahilhabib031/Lab2/pull/new/feature-1
```

```
remote:
```

```
To https://github.com/manahilhabib031/Lab2.git
```

```
* [new branch] feature-1 -> feature-1
```

bugfix/user-auth-error 4 Branches 0 Tags Go to file Add file <> Code

This branch is 3 commits behind main . Contribute

TASK NO 10:

Pull Request and Branch Review (GitHub GUI)

Merged pull req1 #1
manahilhabib031 merged 3 commits into `feature/db-connect...` from `main` now

- `updaed notes.txt` [cc2bf14](#)
- `Add new function to main.py` [8aedabe](#)
- `Merge branch 'feature-2'` [d4bc034](#)

manahilhabib031 merged commit `9f86b29` into `feature/db-connection` [now](#) Revert

Pull request successfully merged and closed
You're all set — the branch has been merged.

Your branches					
Branch	Updated	Check status	Behind	Ahead	Pull request
<code>feature/db-connection</code>	Deleted now		1	0	#2

TASK NO 11:

Detailed Branch Strategy (Develop/Staging)

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (develop)
$ git checkout -b staging
Switched to a new branch 'staging'

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (staging)
$ git push origin staging
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'staging' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/Lab2/pull/new/staging
remote:
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      staging -> staging

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (staging)
$ git checkout -b feature/login-page
Switched to a new branch 'feature/login-page'

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature/login-page)
$ git push origin feature/login-page
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/login-page' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/Lab2/pull/new/feature/login-page
remote:
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      feature/login-page -> feature/login-page

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature/login-page)
$ git checkout -b bugfix/typo-error
git push origin bugfix/typo-error
Switched to a new branch 'bugfix/typo-error'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'bugfix/typo-error' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/Lab2/pull/new/bugfix/typo-error
remote:
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      bugfix/typo-error -> bugfix/typo-error

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/typo-error)
$ |
```



```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/typo-error)
$ git add notes.txt
git commit -m "Add login page feature"
git push origin feature/login-page
On branch bugfix/typo-error
nothing to commit, working tree clean
Everything up-to-date

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/typo-error)
$ git checkout develop
git merge feature/login-page --no-edit
git push origin develop
Switched to branch 'develop'
Already up to date.
Everything up-to-date

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (develop)
$ git checkout staging
git merge develop --no-edit
git push origin staging
Switched to branch 'staging'
Already up to date.
Everything up-to-date

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (staging)
$ git checkout main
git merge staging --no-edit
git push origin main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
Already up to date.
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git add login.py
git commit
fatal: pathspec 'login.py' did not match any files
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git push origin main
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 255 bytes | 255.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/Lab2.git
   ebbb73c..6ec2ea1  main -> main
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git checkout -b develop
fatal: a branch named 'develop' already exists

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git push origin develop
Everything up-to-date

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git checkout -b staging
$ git push origin staging
fatal: a branch named 'staging' already exists
Everything up-to-date

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (main)
$ git checkout -b feature/login-system
$ git push origin feature/login-system

$ git checkout -b bugfix/ui-fix
$ git push origin bugfix/ui-fix
switched to a new branch 'feature/login-system'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/login-system' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/Lab2/pull/new/feature/login-system
remote:
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      feature/login-system -> feature/login-system
switched to a new branch 'bugfix/ui-fix'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'bugfix/ui-fix' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/Lab2/pull/new/bugfix/ui-fix
remote:
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      bugfix/ui-fix -> bugfix/ui-fix
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/ui-fix)
$ git branch -a
  bugfix/typo-error
*  bugfix/ui-fix
  bugfix/user-auth-error
  develop
  feature-1
  feature-2
  feature/db-connection
  feature/login-page
  feature/login-system
  main
  staging
remotes/origin/HEAD -> origin/main
remotes/origin/bugfix/typo-error
remotes/origin/bugfix/ui-fix
remotes/origin/bugfix/user-auth-error
remotes/origin/develop
remotes/origin/feature-1
remotes/origin/feature/db-connection
remotes/origin/feature/login-page
remotes/origin/feature/login-system
remotes/origin/main
remotes/origin/staging
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (bugfix/ui-fix)
$ git checkout feature/login-system
echo "Feature: Added login system" >> notes.txt
git add notes.txt
git commit -m "Add login system feature"
git push origin feature/login-system
Switched to branch 'feature/login-system'
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time you commit.
[feature/login-system 13d8c1d] Add login system feature
1 file changed, 1 insertion(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 318 bytes | 159.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/Lab2.git
6ec2ea1..13d8c1d feature/login-system -> feature/login-system

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature/login-system)
$ git checkout bugfix/ui-fix
echo "Bugfix: Fixed UI issue" >> notes.txt
git add notes.txt
git commit -m "Fix UI issue"
git push origin bugfix/ui-fix
Switched to branch 'bugfix/ui-fix'
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time you commit.
[bugfix/ui-fix 854fcad] Fix UI issue
1 file changed, 1 insertion(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 306 bytes | 153.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/Lab2.git
6ec2ea1..854fcad bugfix/ui-fix -> bugfix/ui-fix
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (develop|MERGING)
$ git checkout staging
git merge develop
git push origin staging
notes.txt: needs merge
error: you need to resolve your current index first
error: Merging is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.
Everything up-to-date

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (develop|MERGING)
$ git checkout main
git merge staging
git push origin main
notes.txt: needs merge
error: you need to resolve your current index first
error: Merging is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.
```

Bonus Task:

Simulated Team Collaboration

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (develop|MERGING)
$ git checkout -b collab
git push origin collab
Switched to a new branch 'collab'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'collab' on GitHub by visiting:
remote:     https://github.com/manahilhabib031/Lab2/pull/new/collab
remote:
To https://github.com/manahilhabib031/Lab2.git
 * [new branch]      collab -> collab
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (collab)
$ git add notes.txt
git commit -m "Added [Your Name] with fun fact"
git push origin collab
[collab 5c556c8] Added [Your Name] with fun fact
 1 file changed, 4 insertions(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 371 bytes | 123.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/Lab2.git
 13d8c1d..5c556c8  collab -> collab
```

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (collab)
$ git add notes.txt
git commit -m "Added manahil with fun fact"
git push origin collab
On branch collab
nothing to commit, working tree clean
Everything up-to-date
```

TASK NO 12:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Lab2 (feature-db-connection)
$ git checkout -b feature-collab
git push origin feature-collab
fatal: a branch named 'feature-collab' already exists
Everything up-to-date
```

EXAM EVALUATION QUESTIONS

Q1 – Advanced Branching & Merge Verification

```

Musfi@DESKTOP-QVT1R80 MINGW64 ~/cc-Lab02 (main)
$ git checkout -b q1-branch
Switched to a new branch 'q1-branch'

Musfi@DESKTOP-QVT1R80 MINGW64 ~/cc-Lab02 (q1-branch)
$ echo "Q1 test line" >> notes.txt
git add notes.txt
git commit -m "Q1: Add test line"
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it
[q1-branch a9b20f9] Q1: Add test line
1 file changed, 1 insertion(+)

Musfi@DESKTOP-QVT1R80 MINGW64 ~/cc-Lab02 (q1-branch)
$ git checkout main
git merge q1-branch
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Updating 8f4c2b0..a9b20f9
Fast-forward
 notes.txt | 1 +
1 file changed, 1 insertion(+)

Musfi@DESKTOP-QVT1R80 MINGW64 ~/cc-Lab02 (main)
$ git log --oneline --graph --all
* a9b20f9 (HEAD -> main, q1-branch) Q1: Add test line
* 8f4c2b0 (origin/main, origin/collab2, collab2, collab) Add teammate's fun fact
* 1e7039d Add teammate's fun fact
* c5bbdfe (origin/collab) Add Musfira's fun fact
* 925d99e (origin/staging, origin/feature/login-page, origin/develop, staging, feature/login-page, develop) Add login feature
* 77d5edc (origin/feature-1) Add new function to main.py
* 4317672 (origin/master, origin/HEAD, master) Merge pull request #1 from Musfira-Farooq/feature/db-connection
/!
* 6dfc7a0 Update notes.txt from db-connection branch
/!
* 3fa2994 (origin/bugfix/user-auth-error) Add third line to notes.txt
* 629ffb1 Update notes.txt with second line
* c20bf90 Update notes.txt and fix README.md
* 52ed362 Update notes.txt with second line
* 16c9d71 Add notes.txt
* f2128e4 Initial commit

```

Q2 – Multi-Stage Workflow Simulation

```

Musfi@DESKTOP-QVT1R80 MINGW64 ~/MultiStageRepo (staging)
$ git log --oneline --graph --all
* bca2fd8 (HEAD -> staging, feature-xyz, develop) Added Feature XYZ
* 0b7668f (master) Initial commit

```

Q3: Collaboration & Conflict Resolution

```

Musfi@DESKTOP-QVT1R80 MINGW64 ~/MultiStageRepo (collaborator2)
$ cat shared.txt
Line from Collaborator 1
Line from Collaborator 2

```

